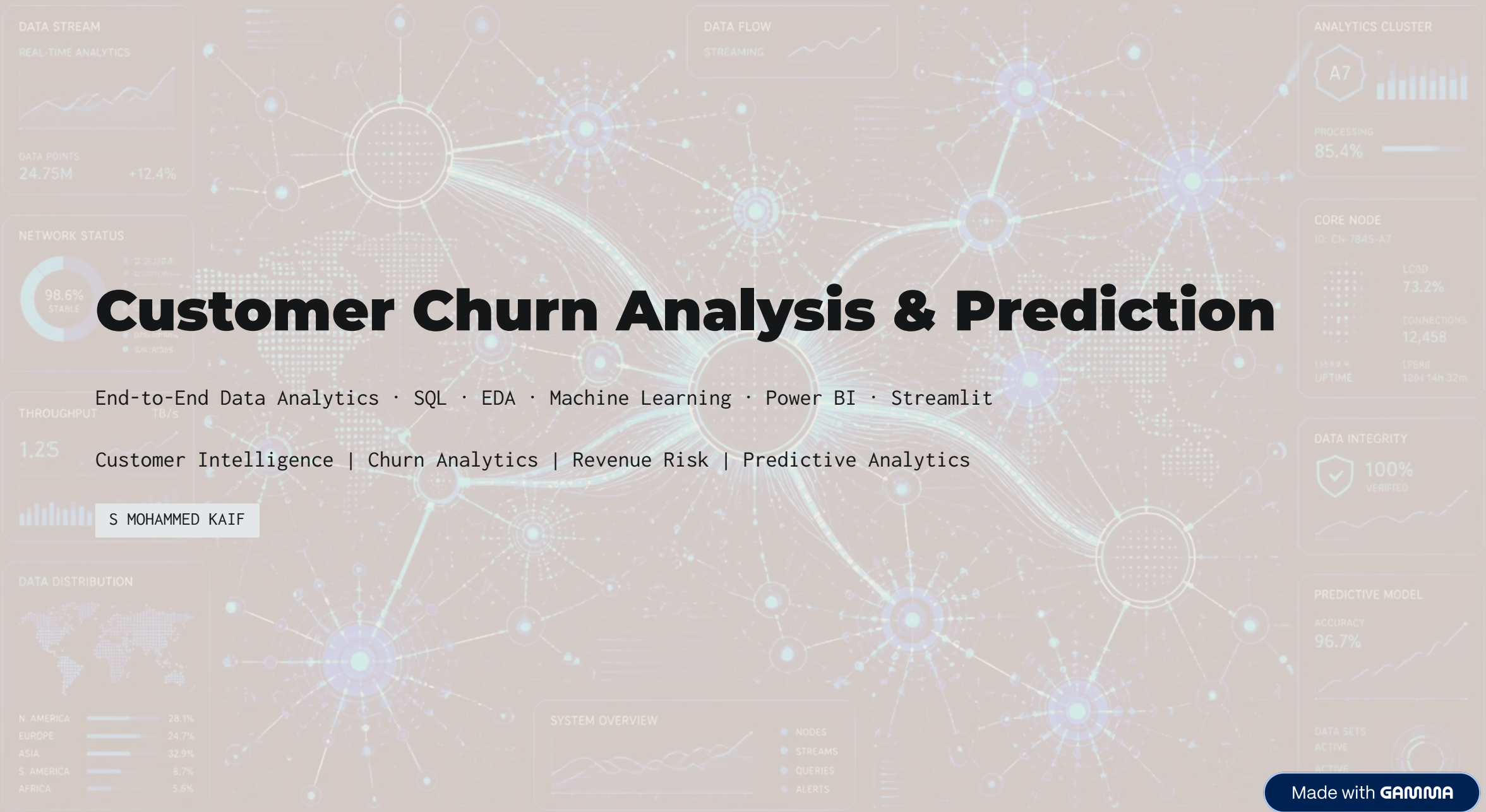




Customer Churn Analysis & Prediction

End-to-End Data Analytics · SQL · EDA · Machine Learning · Power BI · Streamlit

Customer Intelligence | Churn Analytics | Revenue Risk | Predictive Analytics

S MOHAMMED KAIF

Project at a Glance

A complete, production-grade analytics system for a Telecom and Subscription-Based business. This project combines Python data processing, SQL business analysis, machine learning prediction, and interactive dashboards into a unified customer intelligence platform.

Project Type Customer Churn Analytics & Prediction	Domain Telecom / Subscription Business
Approach Data Analytics + SQL + Machine Learning + BI	Output Business Intelligence + Predictive Customer Risk

Technology Stack



Data Processing

Python · Pandas · NumPy



Database

SQLite · SQL



Business Intelligence

Power BI



Geospatial

GeoJSON



Visualization

Matplotlib · Seaborn



Machine Learning

Scikit-Learn



Web App

Streamlit



Notebook

Jupyter

Business Context & The Core Problem

Customers subscribe to a range of services – Phone, Internet, Streaming, Online Security, Online Backup, Device Protection, and Technical Support. Understanding why they leave is critical to protecting revenue and customer lifetime value.

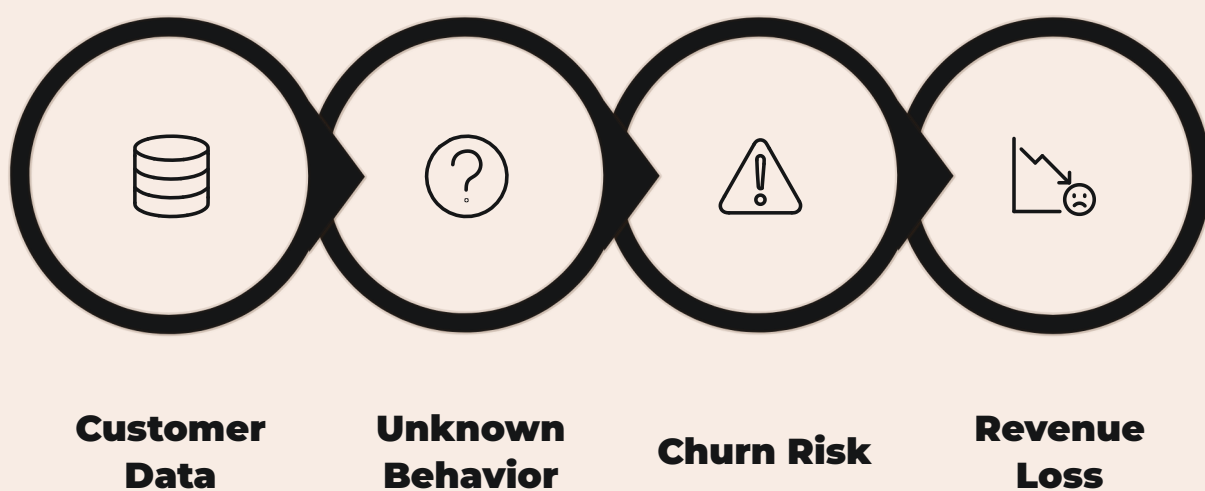
Potential Churn Drivers

- High monthly charges
- Short tenure
- Contract type
- Payment issues
- Service problems
- Poor customer experience
- Support issues
- Customer dissatisfaction

The Central Question

Why are customers leaving? This project addresses three critical business questions:

1. Why are customers leaving?
2. Which customers are most at risk?
3. How much revenue is potentially at risk?



Without structured analytics, churn remains an invisible threat. This project transforms raw customer data into actionable retention intelligence.



Project Objectives & Key Business Questions

This project is structured around a clear set of analytical objectives and business questions that drive every stage of the workflow – from data processing through to machine learning and dashboard delivery.

Project Objectives

Analyze customer demographics and subscription behavior

Study customer tenure and service usage patterns

Identify high-risk customer segments

Analyze contract-level and payment-method churn

Perform SQL business analysis

Build churn prediction models

Create Power BI dashboard and Streamlit web app

Generate actionable business recommendations

Key Business Questions

What is the overall churn rate?

Which customer groups churn most?

Does tenure influence churn?

Which contracts have higher churn?

Which payment methods correlate with churn?

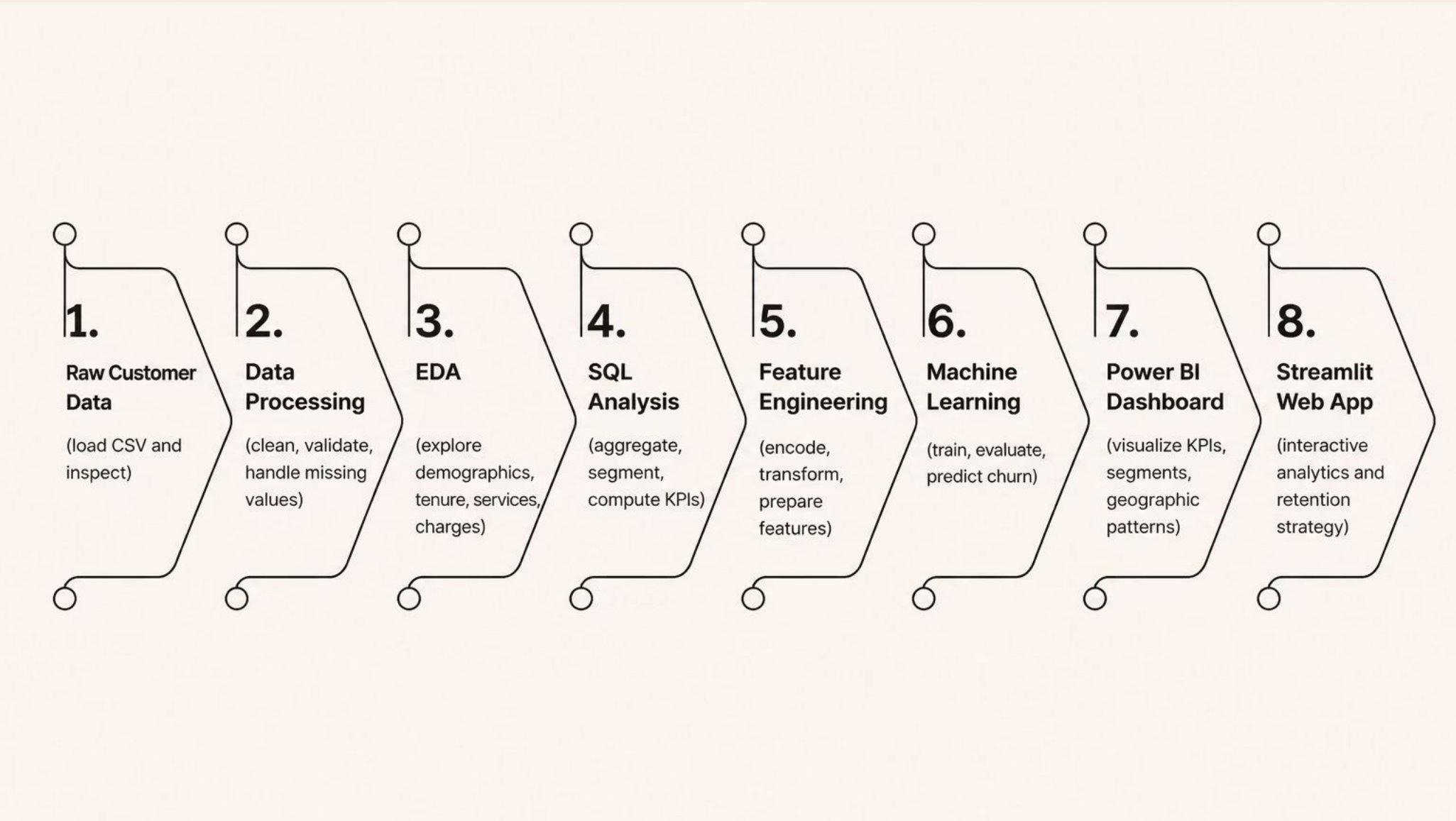
Which states show higher churn?

How do monthly charges relate to churn?

Which customers are predicted to churn?

End-to-End Project Workflow

The project follows a structured, production-grade pipeline – from raw data ingestion through to executive dashboards and a live web application. Each stage is designed to ensure data quality, analytical rigor, and business relevance.



Data Assets

- **CSV:** Churn_Customer_Data.csv, Clean_customer_data.csv, Customer_Data.csv, Customer_Churn_Predictions.csv, Predicted_Churned_Customers.csv
- **Database:** customer_churn.db, customer_data.db
- **Geographic:** india_state.geojson

Data Processing Steps

- Data loading and inspection
- Missing-value analysis
- Duplicate detection and removal
- Data type validation
- Numerical and categorical analysis
- Data consistency checks
- Preparation for analytics

Customer Data Dimensions & EDA

The dataset covers four analytical dimensions spanning customer demographics, service subscriptions, billing information, and lifecycle attributes. Exploratory Data Analysis investigates patterns across each dimension to surface churn signals.

Demographics - Gender - Age - Marital Status - State	Services - Phone Service - Internet Service - Streaming Services - Online Security - Online Backup - Device Protection - Premium Support
Billing - Monthly Charges - Total Charges - Refunds - Extra Data Charges - Long Distance Charges - Total Revenue	Lifecycle - Tenure - Contract Type - Payment Method - Paperless Billing - Customer Status

EDA Focus Areas

Demographics: Gender, age, marital status, and state-level distribution	Tenure: Relationship between customer tenure and churn probability
Services: Internet type, phone service, streaming, and support package uptake	Contracts: Month-to-month vs. long-term contract churn rates
Payment: Payment method correlation with churn behavior	Charges: Monthly and total charge distribution for churned vs. retained customers

Where actual EDA results are unavailable, placeholders such as [INSERT ACTUAL EDA RESULT] are used throughout visualizations and analysis outputs.

Churn Pattern Analysis & SQL Business Intelligence

Churn is analyzed across multiple customer dimensions to identify which segments carry the highest attrition risk. SQL queries on SQLite databases (customer_churn.db, customer_data.db) enable structured business analysis at scale.

Dimension	Churn Rate	Customer Count	Business Interpretation
Contract Type	[ADD ACTUAL CHURN RATE]	[ADD TOTAL CUSTOMERS]	Month-to-month contracts typically show higher churn
Payment Method	[ADD ACTUAL CHURN RATE]	[ADD TOTAL CUSTOMERS]	Electronic check payments often correlate with higher churn
Internet Service	[ADD ACTUAL CHURN RATE]	[ADD TOTAL CUSTOMERS]	Fiber optic users may show elevated churn due to pricing
Tenure Band	[ADD ACTUAL CHURN RATE]	[ADD TOTAL CUSTOMERS]	Newer customers typically carry higher churn risk
Monthly Charges	[ADD ACTUAL CHURN RATE]	[ADD TOTAL CUSTOMERS]	Higher charges may increase churn probability
State	[ADD ACTUAL CHURN RATE]	[ADD TOTAL CUSTOMERS]	Regional patterns reveal geographic churn hotspots

SQL Analytical Categories

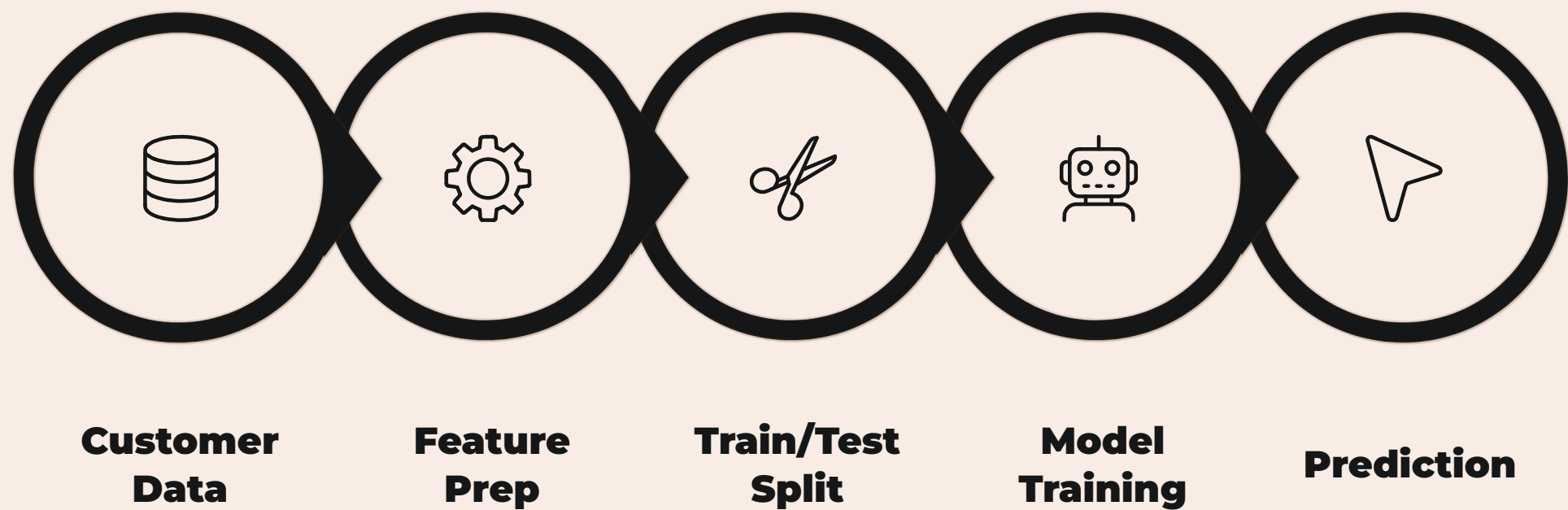
- Customer counts and churn counts
- Churn rate computation by segment
- Revenue aggregation and comparison
- Average monthly charges by group
- Customer segmentation queries
- State-level and contract analysis
- Payment-method breakdown
- Customer status analysis

SQL Pipeline

01
Raw data loaded into SQLite
02
Filter / Group By / Aggregation
03
Business KPI computation
04
Insight extraction

Machine Learning & Churn Prediction

A supervised machine learning model is trained to classify customers as **Churn** or **No Churn**. The pipeline transforms raw customer features into a trained, evaluated prediction model capable of scoring individual churn risk.



The model target is binary: **CHURN** vs. **NO CHURN**. Features are encoded, scaled, and split for training and testing. Model performance is evaluated using standard classification metrics.

Prediction Output Files

- Customer_Churn_Predictions.csv
- Predicted_Churned_Customers.csv

Each customer record is assigned a predicted risk category and a recommended retention action priority.

Conceptual Risk Framework

Low Risk Monitor periodically	Medium Risk Targeted engagement
High Risk Immediate retention action	

📄 Model algorithm, accuracy, ROC-AUC, and feature importance values should be populated with actual project results: **[INSERT ACTUAL MODEL USED]**, **[INSERT ACTUAL PERFORMANCE]**.

Power BI Dashboard, Streamlit App & Geographic Analysis

The analytical outputs are delivered through two interactive layers: an executive Power BI dashboard and a Streamlit web application, enabling stakeholders to explore churn insights without technical dependencies.



Power BI Dashboard

`Churn_Analysis_BI_Dashbord.pbix`

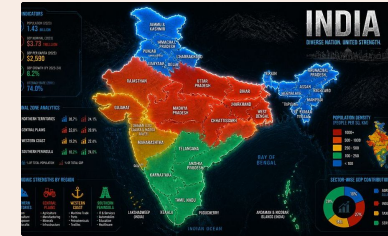
- Customer KPIs and churn rate
- Customer status and revenue
- Services, contracts, and payment methods
- Geographic churn distribution
- Customer segment breakdowns



Streamlit Web Application

`Stream_app.py` ·
`master_churn_dashboard.py`

- KPI cards and customer analytics
- Churn and revenue analysis
- Interactive filters and charts
- Customer segmentation views
- Prediction analysis interface



Geographic Churn Analysis

`india_state.geojson`

- Customer distribution by state
- Churn concentration mapping
- Revenue by geographic region
- High-risk geographic segments
- Regional pattern identification

Business Impact, Deliverables & Portfolio

This project transforms raw customer data into a complete business intelligence system – from exploratory analysis and SQL queries to machine learning predictions, interactive dashboards, and actionable retention strategies.

27%

Customer Churn Rate

73%

Retention Rate

6418

Total Customers

1732

Churned Customers

17.34 mths

Average Tenure

3411960.58

Revenue at Risk

Business Recommendations

- 1

Prioritize high-risk customers for immediate retention outreach
- 2

Encourage longer-term contracts to reduce month-to-month churn
- 3

Improve onboarding and customer support experience
- 4

Build proactive retention campaigns using predictive analytics
- 5

Monitor churn KPIs continuously via Power BI and Streamlit

Project Deliverables

- 01-Churn-Analysis-proj-for-EDA-Data-Processing.ipynb
- 02-Churn-SQL-Dashbord-Analysis.ipynb
- 03-Churn-Prediction-Analysis.ipynb
- CSV datasets and SQLite databases
- Power BI dashboard (.pbix)
- Streamlit web application
- Geospatial analysis (.geojson)

Explore the Project

- View GitHub P...

LinkedIn Pr...

Streamlit W...
- Power BI Dashboard

"Turning Customer Data into Business Intelligence" – S Mohammed Kaif

Thank you for reviewing this project. This end-to-end analytics system demonstrates proficiency in data processing, SQL, machine learning, business intelligence, and full-stack data application development.